TABLE I
Comprehensive Experimental Extensions: Quantum Control, Late Fusion, Label Noise Sensitivity, and Theoretical Ceiling

Module	Model Configuration / Setup	Primary Metric	95% CI / Key Empirical Finding
Part A: Quantum Control	Classical RBF Kernel SVC (4-PCA, $\gamma = 0.25$)	QWK = **0.7763**	$[0.7570, 0.7950]$ (Superior test generalization)
	Quantum Fidelity Kernel SVC (4-qubit ZZFeatureMap)	QWK = 0.7109	$[0.6895, 0.7312]$ (Inferior downstream accuracy)
	Classical Kernel-Target Alignment (KTA)	KTA = 0.1170	Downstream generalization superior (ΔQWK = $+0.065$)
	Quantum Kernel-Target Alignment (KTA)	KTA = **0.1957**	Higher Hilbert alignment, but lower test generalization
Part B: Late Fusion ($B = 2000$)	Classical ElasticNet LR (Pooled Out-of-Fold)	Macro-F1 = **0.5590**	QWK = **0.8099**
	Quantum Fidelity Kernel SVC (Pooled Out-of-Fold)	Macro-F1 = 0.4195	QWK = 0.7109
	Weighted Late Fusion (0.75C + 0.25Q)	Macro-F1 = 0.5109	QWK = 0.8042
	Stacking Meta-Classifier (Multinomial LR on OOF)	Macro-F1 = 0.5357	QWK = 0.7981
	Paired Bootstrap Delta (Δ = Fusion − Classical)	ΔF1 = -0.0481	95% CI: $[-0.0918, -0.0046]$ ($p < 0.05$ degradation); ΔQWK spans 0
Part C: Label Sensitivity	Male-Trained Model on Male Target	Accuracy = 54.01%	QWK = 0.8029
	Male-Trained Model on Midpoint Target	Accuracy = 51.87%	QWK = **0.7931** (Robust surrogate generalization)
	Sensitivity Across Sex Assumptions	125/374 Switch	Boundary shifts without model collapse
	Filtered Concordant Cohort ($N = 249$)	Accuracy = 50.06%	QWK = 0.8029 (Creatinine-selection bias; rejected)
Part D: Ceiling Check	Primary Leakage-Free Model (LR ElasticNet)	QWK = 0.8029	Macro-F1 = 0.5414
	Upper-Bound Ceiling Model (S_{cr} + Age Allowed)	QWK = **0.8802**	Macro-F1 = **0.6612** (Accuracy = 65.26%)
	Relative Ceiling Attained by Leak-Free Model	93.2% of Maximum Theoretical Attainable Ceiling on Cohort	